

Effects of Ankle Tendon Electrical Stimulation on Detection Threshold of Redirected Walking

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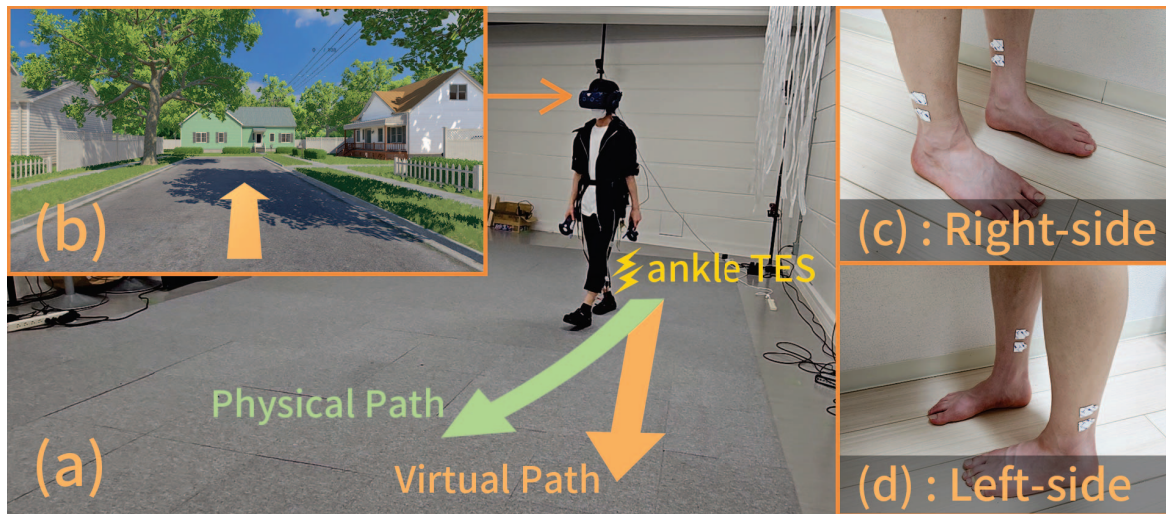


Figure 1: (a): Participants performed redirected walking with ankle tendon electrical stimulation applied. (b): VR view of the experimental system. (c): Electrode position of the stimulation site in the right-side (RS) condition. (d): Electrode position of the stimulation site in the left-side (LS) condition.

ABSTRACT

Redirected walking (RDW) allows users to explore virtual spaces larger than physical spaces while preserving a natural walking sensation. Expanding the range of visual manipulation gains that can be applied without causing discomfort is necessary to apply RDW in practice. Ankle tendon electrical stimulation (TES) has the potential to expand the range by inducing body tilt sensation and body sway. Therefore, in this study, we proposed a locomotion method that applies ankle TES to RDW. In the experiment, we evaluated how ankle TES affects the detection threshold (DT), which is the maximal gain at which visual manipulation is not noticed. The results indicated that the DT was expanded when TES was applied to induce the body tilt sensation in the same direction as the RDW's visual manipulation. Specifically, the pooled mean of the DT was expanded by more than 18%.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—

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Interaction devices—Haptic devices

1 INTRODUCTION

Redirected Walking (RDW) [3] is a promising VR locomotion method for Virtual Reality (VR) applications that require a high immersion. It is achieved by the RDW's ability to generate a natural walking sensation stemming from its actual walking motions. Moreover, it does not require any additional hardware as long as a set of HMD is provided. To realize RDW with little discomfort or sickness, it is necessary to convert the user's position/posture within the range not perceived by the user. This imperceptible conversion amount is termed the detection threshold (DT). A larger DT allows the user to explore a vaster virtual space within a constrained physical space. Therefore, expanding the DT is an important research topic in the field of RDW. In this study, we hypothesized that ankle tendon electrical stimulation (TES), which is known to induce a body tilt sensation and body sway, could be applied to RDW to expand the DT. We investigated the effect of ankle TES on the RDW in 26 experimental participants.

2 PROPOSED METHOD

TES evokes a reflex known as the Golgi tendon reflex, which causes the tendons to elongate. A previous study has reported that electrical stimulation of the tibialis anterior muscle tendon, Achilles tendon, peroneus longus muscle tendon, and flexor digitorum longus muscle tendon, which are located at the ankle, induces a body tilt sensation and a body sway in the direction opposite to the stimulation site [5]. For example, ankle TES of the RS condition (Fig.1 (c)) causes a leftward body tilt sensation and body sway, while ankle TES of the

LS condition (Fig.1 (d)) causes a rightward body tilt sensation and body sway. Based on this finding, we hypothesized that applying ankle TES to RDW would naturally cause a sense of physical path bending due to its body swaying effect and expand the DT of RDW.

3 EXPERIMENT

This experiment aims to investigate the effect of ankle TES on the DT of RDW. This experiment was approved by the local ethics committee of our organization.

We recruited 26 experimental participants consisting of 12 males and 14 females, with a mean age of 25.2. Figure 1 (a) depicts the experimental environment. The experiment was conducted in a room with an area of at least 4m square. Experimental participants experienced RDW with curvature gain while wearing a backpack PC (HP VR Backpack G2) and a head-mounted display (HTC Vive Pro Eye) and connected an electrical stimulator to both ankles. This electrical stimulator uses a constant-current circuit and can apply a current of up to 4mA. In this experiment, an 80Hz bipolar square wave with a duty ratio of 50% was used, and the intensity was adjusted for each experimental participant.

To investigate the effect of ankle TES on the DT of RDW, experimental participants experienced RDW with different combinations of gain and electrical stimulation conditions: nine gain conditions (g_C : -0.2, -0.1, -0.05, -0.025, 0, +0.025, +0.05, +0.1, +0.2), three electrical stimulation conditions (LS: stimulation to the outside of the left foot and inside of the right foot, RS: stimulation to the inside of the left foot and outside of the right foot, NONE: no stimulation. Stimulation sites in the LS and RS conditions are depicted in Fig. 1 (c) and (d), respectively). Four trials were performed for each combination, and 270 ($= 9 \times 3 \times 10$) trials were performed in total.

First, the experimental participants adjusted the electrical stimulation intensity for the each electrical stimulation conditions so that the electrical stimulation was at the maximum pain-free intensity. In each walking task, the experimental participants walked 4m with curvature gain applied. Then, based on a previous study that measured DT using pseudo two-alternative forced choice method [4], participants answered the question, "Did you feel walking left or right?", with choices of right or left in each trial (pseudo-2AFC task). Then, the participants moved to the starting point of the next task. The above walking task was performed 270 times with several breaks in between.

4 RESULT & CONCLUSION

To evaluate the DT of RDW, in accordance with the procedures in the previous study [4], the responses to the pseudo-2AFC task were fitted to a cumulative Gaussian function using Psignifit4 [2]. The curvature gains at which the right response rate to the pseudo-2AFC task was 25% and 75% were defined as the leftward DT and rightward DT, respectively. The box plots of the leftward DT and rightward DT obtained under each electrical stimulation condition are described in Fig. 2. To examine the differences between the electrical stimulation conditions, we performed a paired t-test with the Bonferroni correction. According to the test results, there was a significant difference between the LS and NONE conditions for the rightward DT ($p = 0.0147, d = 0.5388$), indicating that the rightward DT is expanded significantly by the LS condition's ankle TES. As for the leftward DT, there was a marginally significant difference between the NONE and RS conditions ($p = 0.0316, d = 0.4673$), indicating that the leftward DT is expanded by the RS condition's ankle TES. No significant differences were found in the other comparisons. As for the pooled mean of DT, the pooled means of the rightward DT were 0.089 in the LS condition, 0.071 in the NONE condition, and 0.071 in the RS condition, while those of the leftward DT were -0.082 in the LS condition, -0.081 in the NONE condition, and -0.096 in the RS condition. The pooled mean of the rightward DT increased by 18.1% in the LS condition compared to the NONE

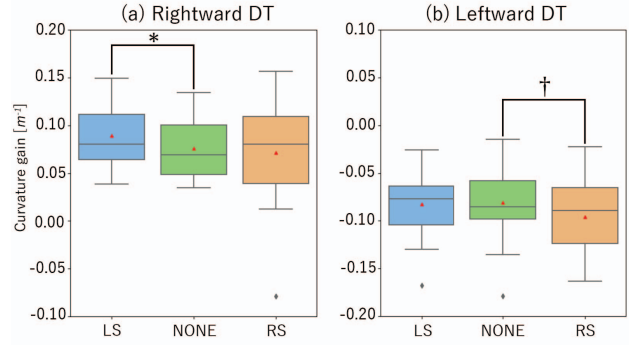


Figure 2: (a): Box plot of the rightward DT in each electrical stimulation. (b): Box plot of the leftward DT in each electrical stimulation. Symbols of * and † indicate a significant difference and a marginally significant difference, respectively. Red markers represent means, and gray markers represent outliers. The outliers are defined as the data located at distances greater than 1.5 times the interquartile range from the first or third quartile.

condition, and the pooled mean of the leftward DT increased by 18.3% in the RS condition compared to the NONE condition.

This result indicates that the LS condition's ankle TES expands rightward DT, while the RS condition's ankle TES expands leftward DT. In a typical VR system using RDW, the rightward curvature gain and the leftward curvature gain are used differently depending on the situation. Therefore, the virtual space expansion capability of the entire RDW system could be improved by applying the RS condition's ankle TES when the leftward curvature gain is used and applying the LS condition's ankle TES when the rightward curvature gain is used.

Two participants commented that they felt comfortable when the direction of the body tilt sensation induced by ankle TES matched the direction of the curve of the path in the physical space. This comment suggests that ankle TES may allow users to walk comfortably even with a perceptible curvature manipulation. Therefore, it is worth evaluating the applicability [1], an index to assess the comfort and usability of VR locomotion methods.

ACKNOWLEDGMENTS

This study was supported by JSPS KAKENHI (22K17929, 22H03628, and 21H04883), JST SOUHATSU (JPMJFR226M), and Leave a Nest Research Fund.

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